Q.1 {\text{Which of the following sets is countable and also a Borel set with measure zero?}}

A. {{The set of irrational numbers}}
B. {{The interval [0,1]}}
C. {{The set of real numbers}}
D. {{The set of natural numbers}}

Q.2 In \left[0,1\right) , the sum modulo 1,x\widehat{+}y of \frac{3}{2} and \frac{1}{2}=_____.

A. {1}{2}
B. is not defined at all
C. 3
D. 1

Q.3 Which of the following is a property of Borel sets?

A. The complement of a Borel set is always an open set.
B. Borel sets are always finite.
C. Borel sets cannot contain uncountably many elements.
D. Every Borel set is measurable.

Q.4 {\text{Which operation generates }}{G_\delta }{\text{ Borel sets from open sets?}}

A. Countable intersection
B. Countable complementation
C. Countable union
D. Countable negation

Q.5 A set E is a Borel set if it can be expressed as_____.

A. countable union of open (closed) sets
B. countable intersection of open (closed) sets
C. neither countable union nor intersection of open or closed sets
D. either countable union or intersection of open or closed sets

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Q.6 Let M be a measurable space (a sigma algebra of measurable sets) then a function f:E\longrightarrow %TCIMACRO{\U{211d} }% %BeginExpansion \mathbb{R} %EndExpansion on \varnothing \neq E\in M given as \left\{ x\in E:f(x)>a\right\} is measurable if \forall a,----------under f is measurable set.

A. inverse image of (a,)
B. image of (a,)
C. image of (x,)
D. inverse image of (a,)

Q.7 Those countable sets which are Borel have _____ measure.

A. no
B. positive
C. infinite
D. zero

Q.8 Suppose f is a real valued identically constant function f(x)=\beta , then \left\{ x:f(x)>\alpha \right\} =% %TCIMACRO{\U{211d} }% %BeginExpansion \mathbb{R} %EndExpansion. And if \beta >\alpha, then_____.

A. f is 1-1

B. f is measurable

C. f~is not measurable

D. f is bijective

Q.9 {\text{A singleton set }}\{ x\} {\text{ can be written as:}}

A. _{n = 1}^ {({{1}{n},{{ }}{1}{n}})}

B. _{n = 1}^ {({x - {1}{n},{{ }}x + {1}{n}})}

C. _{n = 1}^ {({x - {1}{n},{{ }}x + {1}{n}})}

D. _{n = 1}^ {({ - {1}{n},{{ }}{1}{n}})}

Q.10 A(an)_____of Lebesgue measure m to the class of Borel sets % \mathit{B} is said to be Borel measure space.

A. bijection

B. extension

C. restriction

D. kernel
